

Containers and Kubernetes TASK – 1

- 1. Install Docker Desktop or Docker Engine on your local machine and verify the installation by running the command `docker --version` in your terminal.**

Ans :

- Go to the Docker website.
 - Download Docker Desktop for your computer.
 - Install Docker Desktop
 - Open the downloaded file.
 - Click Install.
 - Follow the instructions on the screen.
 - Restart your computer if Docker asks you to.
 - Open Docker Desktop.
 - Wait until Docker is running.
 - Open Command Prompt (Windows) or Terminal (Mac/Linux).
 - Type this command:
 - `docker --version`
 - Press Enter.
 - You should see something like:
 - Docker version 28.x.x
- 2. Pull the official Nginx Docker image from Docker Hub using the command `docker pull nginx` and list all downloaded images with `docker images`.**

Ans :

- Open Docker Desktop.
 - Make sure Docker is running.
 - Open Command Prompt or Terminal.
 - Download the Nginx image by typing:
 - `docker pull nginx`
 - Press Enter.
 - Wait for the download to finish. You should see something like:
 - Status: Downloaded newer image for nginx:latest
 - List all downloaded Docker images by typing:
 - `docker images`
 - Press Enter.
 - You should see something similar to:
 - | REPOSITORY | TAG | IMAGE ID | CREATED | SIZE |
|------------|--------|-----------|---------|------|
| nginx | latest | xxxxxxxxx | ... | ... |
 - The important part is that you can see:
 - `nginx latest`
- 3. Start a new Nginx container using the command `docker run -d -p 8080:80 --name mynginx nginx`, then open your browser and check if you can access the default Nginx welcome page at <http://localhost:8080>.**

Ans :

- Make sure Docker Desktop is open and running.
- Open Command Prompt or Terminal.
- Run this command:
- `docker run -d -p 8080:80 --name mynginx nginx`
- Press Enter.

- You should get a long container ID, something like:
- 8f3a2c1d5e...
- This means the container has started successfully.
- Check that the container is running by typing:
- `docker ps`
- You should see mynginx in the list.
- Open your web browser (Chrome, Edge, Firefox, etc.).
- In the address bar, enter:
- `http://localhost:8080`
- Press Enter.
- You should see the “Welcome to nginx!” page.
- Commands you need
- First:
- `docker run -d -p 8080:80 --name mynginx nginx`
- Then check:
- `docker ps`
- Finally, open:
- `http://localhost:8080`

**4. Create a Docker volume named zomato_data using docker volume create and attach it to a new container running the httpd (Apache) image so that any data written to /usr/local/apache2/htdocs inside the container is stored persistently.

Hint: Use the -v flag with docker run to mount the volume.**

Ans :

- Open Docker Desktop and make sure it is running.

- Open Command Prompt or Terminal.
- Create a volume named zomato_data:
- `docker volume create zomato_data`
- You should see:
- `zomato_data`
- This means the volume was created successfully.
- Check the volume:
- `docker volume ls`
- You should see:
- DRIVER VOLUME NAME
- `local zomato_data`
- Start a new Apache container and attach the volume:
- `docker run -d --name myapache -v zomato_data:/usr/local/apache2/htdocs httpd`
- The command means:
 - `-d` → runs the container in the background.
 - `--name myapache` → gives the container the name `myapache`.
 - `-v zomato_data:/usr/local/apache2/htdocs` → attaches the `zomato_data` volume to Apache's website folder.
 - `httpd` → uses the Apache HTTP Server image.
- Check that the Apache container is running:
- `docker ps`
- You should see `myapache` in the list.

- Run these one by one:
- `docker volume create zomato_data`
- `docker volume ls`
- `docker run -d --name myapache -v zomato_data:/usr/local/apache2/htdocs httpd`
- `docker ps`

5. Compare containers and virtual machines by listing 3 benefits of Docker containers over traditional VMs, using examples from apps like Zomato, Instagram, or Flipkart.

Ans :

Benefit	Docker Containers	Traditional VMs	Example
1. Faster startup	Containers can start in seconds because they share the host OS kernel.	VMs need to boot a complete operating system, so they take longer.	Zomato could quickly start new containers when food-order traffic suddenly increases.
2. Uses fewer resources	Containers are lightweight and use less RAM and CPU.	Each VM requires its own operating system, using more resources.	Instagram could run many small services on the same server using containers.
3. Easy to deploy and scale	The same Docker image can be deployed consistently on	Setting up and managing complete VM environments is generally heavier.	Flipkart could create additional containers during a big sale to handle

Benefit	Docker Containers	Traditional VMs	Example
	different servers.		increased shopping traffic.